

Droplet 2020

Special Relativity & Black Holes

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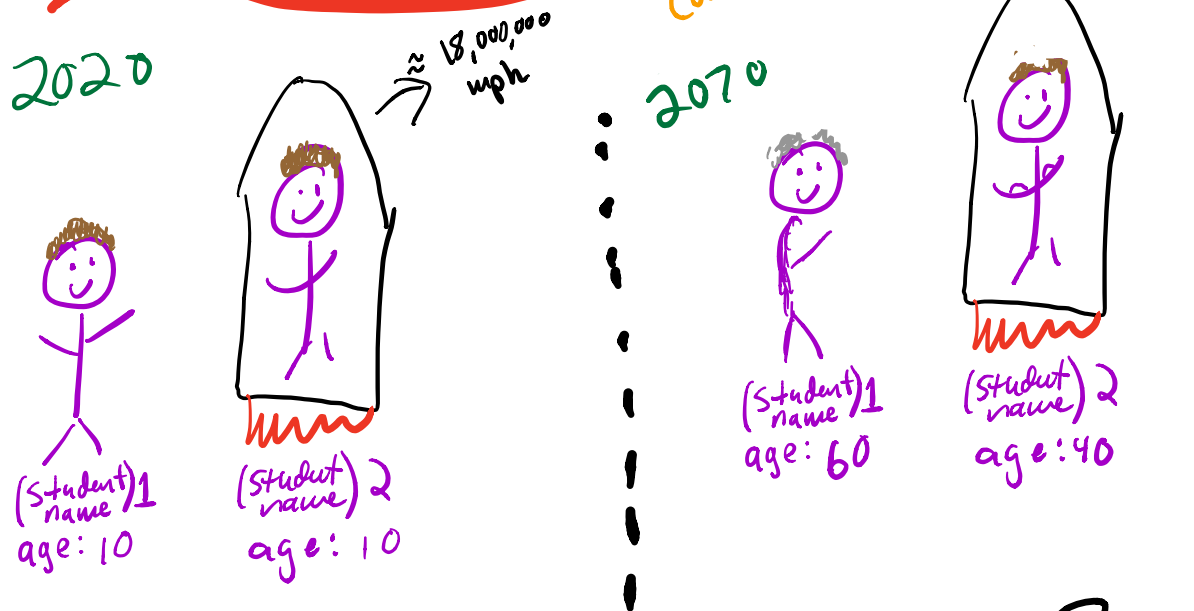
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1.) What's special about Special relativity? ①

Twin Paradox (Conceptual)



Q: Why does this happen?

Because the Universe is 4D!
4th Dimension is time

3D we see: space

up/down

forward/backward

Left/right

Time! (1D)

= 4D Spacetime

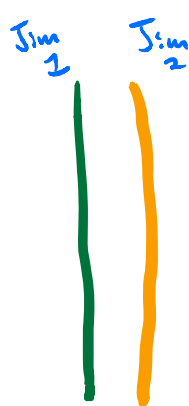
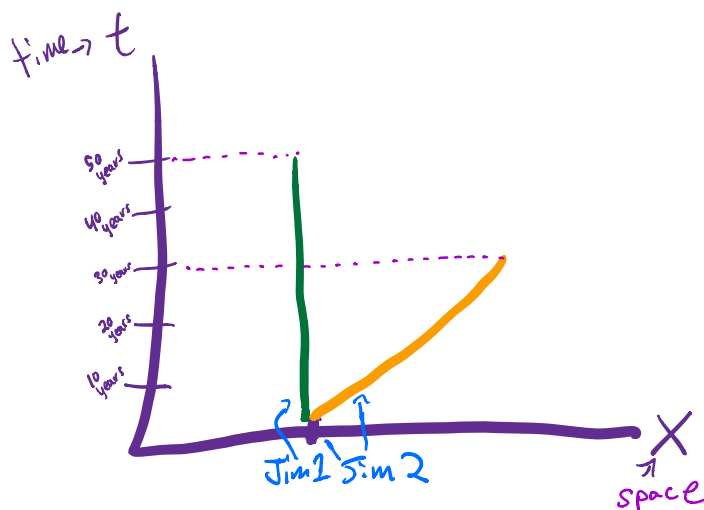
This block explains the concept of 4D spacetime. It lists the three dimensions of space we can see: up/down, forward/backward, and left/right. It then adds the fourth dimension, time, represented by a clock icon. The equation shows that these four dimensions combined equal 4D spacetime.

We treat time as another direction to move in. (2)

We say since the traveling twin moves more through space he moves less through time.

An important property is that 4D spacetime distance traveled remains the same.

Lets see this (slightly inaccurate diagram but good for current purpose)

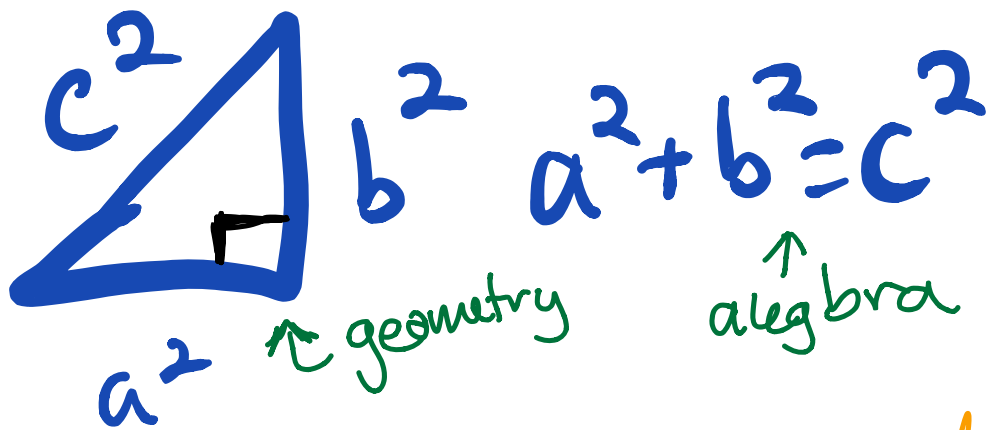


We see the length is the same
Since Jim 2 moved through space he moved less through time

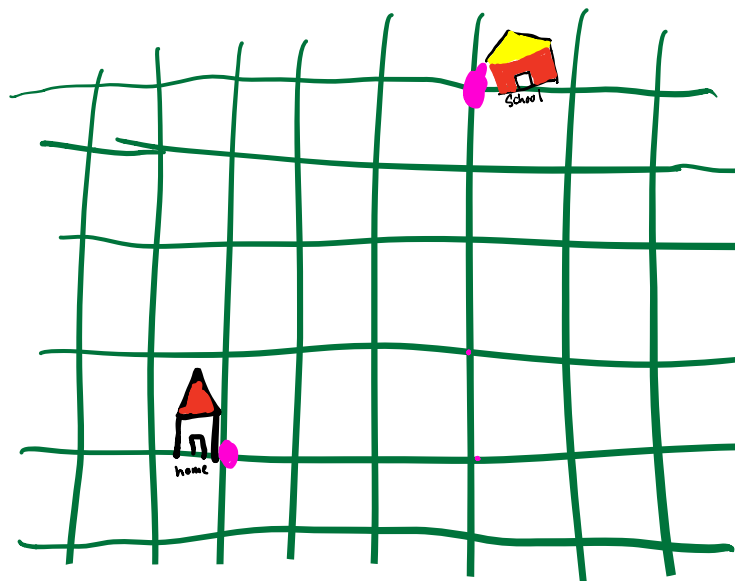
At low speeds (≈ 60 mph) this still happens but the effect is negligible

2.) How do we make a ③
4 dimensional geometry that
involves a time dilation?

First lets review :

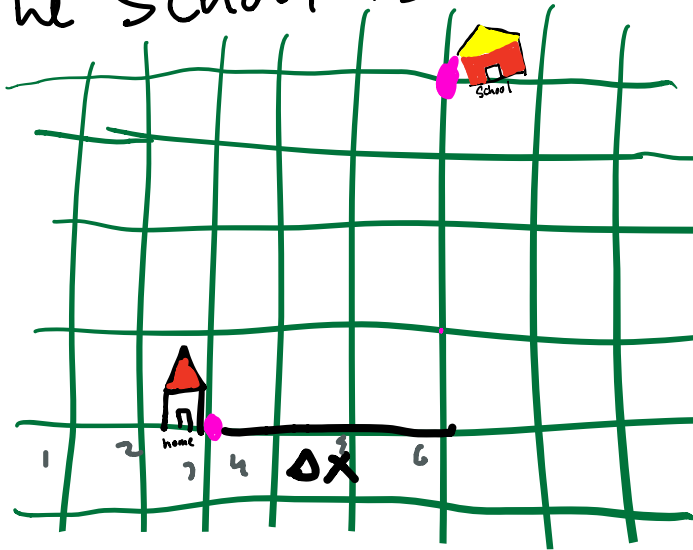


We can use this every day



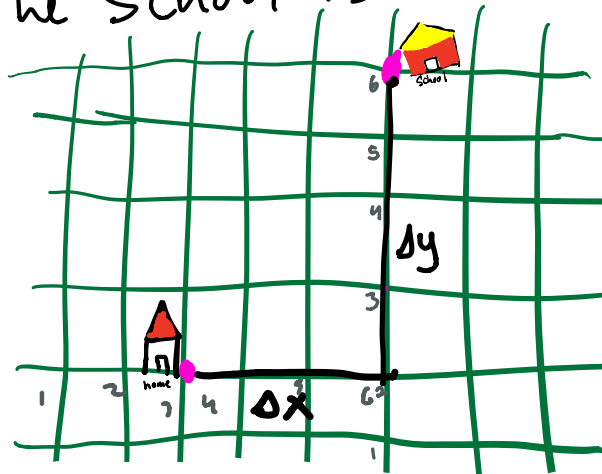
The School is 3 blocks east

(4)



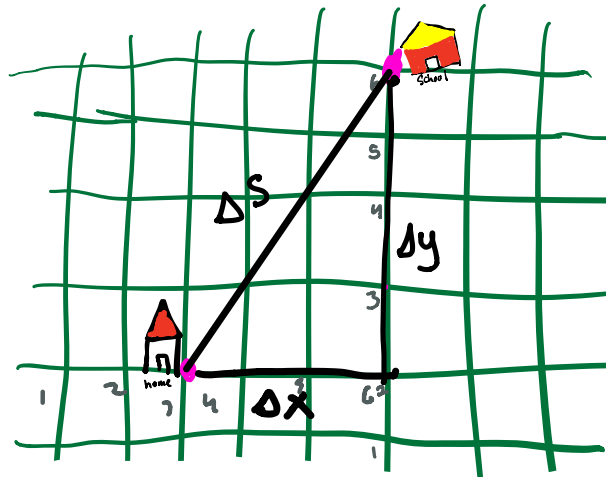
In our language we call the total distance traveled in the east direction " Δx " $\Delta x = 6 - 3 = 3$

The School is 4 blocks North



In our language we call the total distance traveled in the North direction " Δy " $\Delta y = 6 - 2 = 4$

We call the shortest distance between home and school " ΔS " ⑤



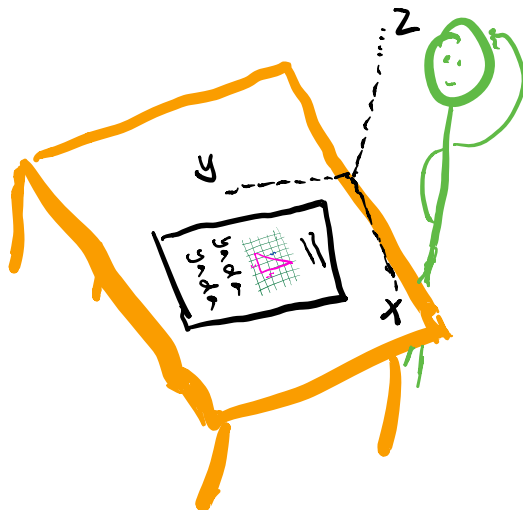
Using $a^2 + b^2 = c^2$ we find $3^2 + 4^2 = 5^2$!

In the 2D paper world we say

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2$$

$\nearrow$
east/west
 $\uparrow$
north/south

But we do not live on paper...



The World our eyes see:

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$



But wait! Didn't all this twin paradox business tell us we live in 4 dimensions?

Naively we will write

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 + (\Delta t)^2$$

time
↓

Einstein showed the geometry of our universe is:

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c \Delta t)^2$$

↑
minus
sign!

* Comments *

• The obvious minus sign.
We will see its importance later

One way of understanding is that time is an "imaginary dimension" $(i\Delta t)^2 = -\Delta t^2$

• The c coming in

c means speed of light.
"How fast light travels"

There is important physics here
for now we can see its necessity with

$$\Delta S = \Delta x = \Delta y = \Delta z = \text{meters / feet / miles ect. ...}$$

$$\Delta t = \text{seconds / hours / years ect. ...}$$

How do you normally think of
Speed? Miles per hour maybe?

What does "miles per hour" mean?

$$v = \frac{\text{Miles}}{\text{hour}} = \frac{60 \text{ miles}}{1 \text{ hour}} = 60 \text{ mph} \quad \frac{90 \text{ miles}}{3 \text{ hours}} = \frac{30 \text{ miles}}{1 \text{ hour}} = 30 \text{ mph}$$

We see $v \times (\text{time}) = \text{distance}$

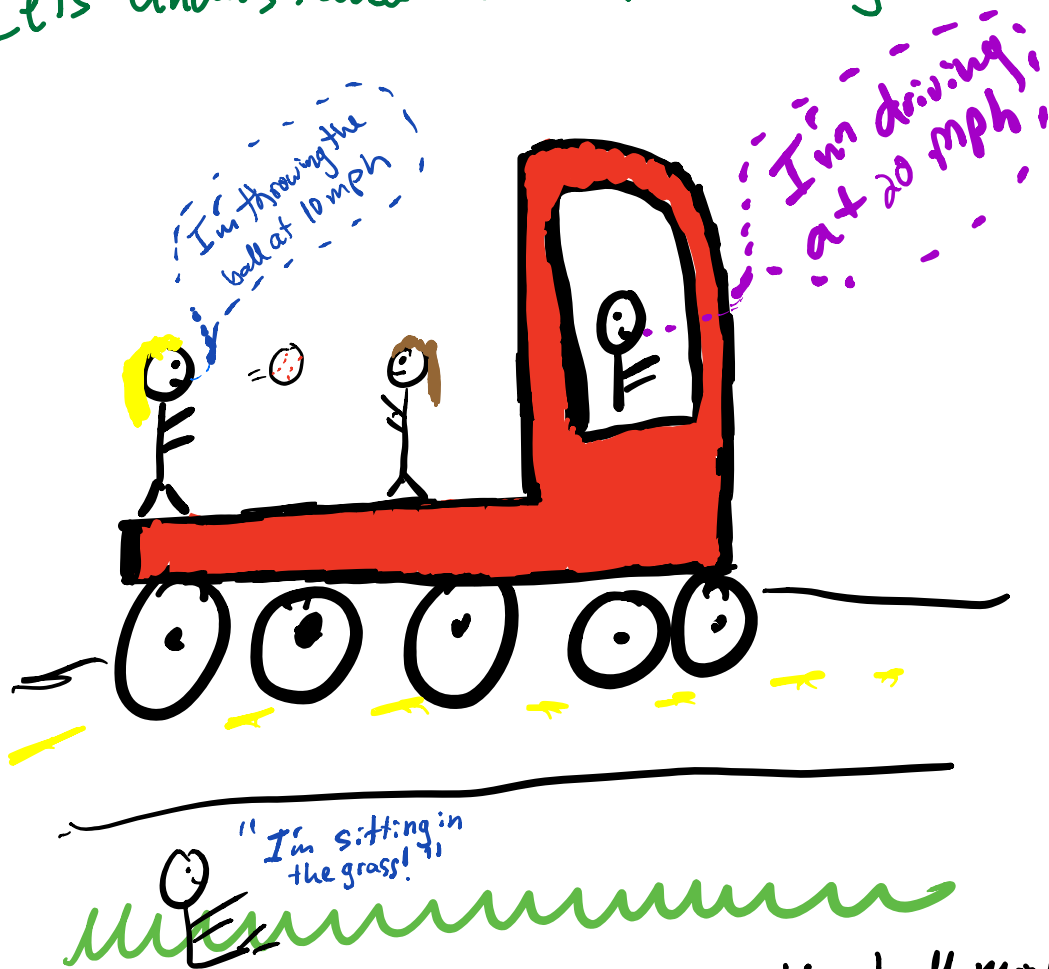
in our notation $v \Delta t = \Delta S$

Now take $v = c$

3.) Why the speed of light? ⑧

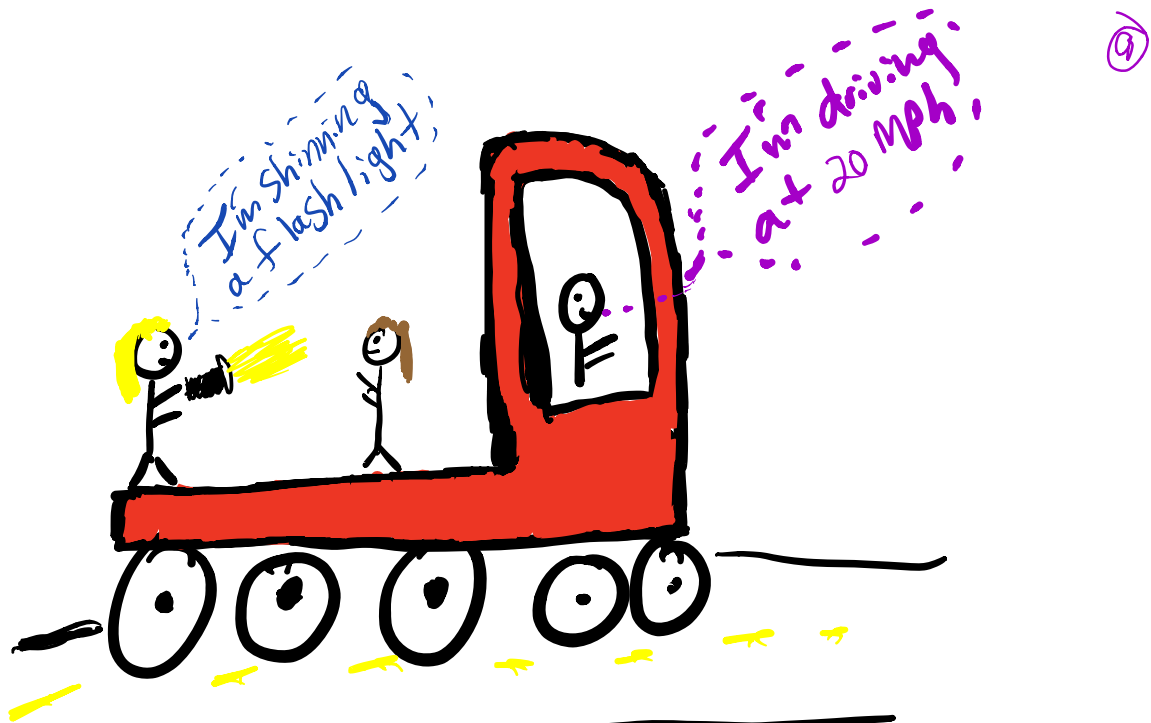
According to the laws of electricity & magnetism light moves the same speed for everyone! (see Michelson-Morely Experiment)

Lets understand this fascinating fact



Q: what speed does the man see the ball move?

$$10 \text{ mph} + 20 \text{ mph} = 30 \text{ mph}$$



Q: What speed does the man see the beam of light move? $C + 20\text{mph} = C$?

This is obvious not the right way to look at things. It turns out for this experimentally observed principle to hold we must live in

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c\Delta t)^2$$

A consequence of this is that (10)
time dilates, length contracts,
and nothing can go faster than light

A common thing to do in physics is
to rewrite our equation to see things
better

Do not worry if you have trouble
following the steps. The main thing
you see the equation on page ...

$$\frac{(\Delta S)^2}{c^2(\Delta t)^2} = \frac{(\Delta x)^2}{c^2(\Delta t)^2} + \frac{(\Delta y)^2}{c^2(\Delta t)^2} + \frac{(\Delta z)^2}{c^2(\Delta t)^2} - \frac{c^2(\Delta t)^2}{c^2(\Delta t)^2}$$

$$\text{Speed in } x: v_x = \frac{\Delta x}{\Delta t} \quad \text{Speed in } y: v_y = \frac{\Delta y}{\Delta t} \quad \text{Speed in } z: v_z = \frac{\Delta z}{\Delta t}$$

$$\frac{(\Delta S)^2}{c^2(\Delta t)^2} = \frac{v^2}{c^2} - 1 \quad \text{where} \quad v^2 = v_x^2 + v_y^2 + v_z^2$$

Multiply both sides by $-(\Delta t)^2$

$$-(\Delta S)^2 = \Delta t \left(1 - \frac{v^2}{c^2} \right)$$

We can look at the units

$$\Delta S / c = \text{distance} / (\text{distance} / \text{time}) = \text{time}$$

(11)

Sticking to our convention: $-c^2(\Delta\tau)^2 = (\Delta S)^2$

$$(\Delta\tau)^2 = -(\Delta S)^2 / c^2$$

$$(\Delta\tau)^2 = (\Delta t)^2 \left(1 - \frac{v^2}{c^2}\right)$$

time of moving twin

time of twin on earth

speed of moving twin

A more indepth approach involves realizing space time distance does not change.

$$-c^2\Delta\tau^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2$$

P.O.V. of moving twin who sees himself at rest

P.O.V. of twin on earth who sees the other twin moving.

But lets calculate something.

I am reading the newspaper for 10 mins at home on my couch. How long does someone moving half the speed of light see me reading the news paper?

My time: $\Delta t = 10 \text{ mins}$ travels Speed: $v = \frac{1}{2}c$

$$(\Delta\tau)^2 = (\Delta t)^2 \left(1 - \frac{v^2}{c^2}\right) = 100 \left(1 - \frac{1}{4}\right) = 75$$

$$(\Delta\tau)^2 = 75 \Rightarrow \Delta\tau = 8.66 \text{ mins}$$

General Relativity

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What is gravity?

We learn in school Newton discovered that "mass attracts mass"

The equation is $F = \frac{G m_1 m_2}{r^2}$

Force moving them together.

two massive objects

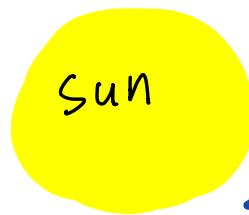
However in 1917 an experiment was done to show this is wrong!

Light has no mass but yet gets attracted by gravity

We see

Newton said:

$$F = \frac{G m_{\text{sun}} 0}{r^2} = 0$$

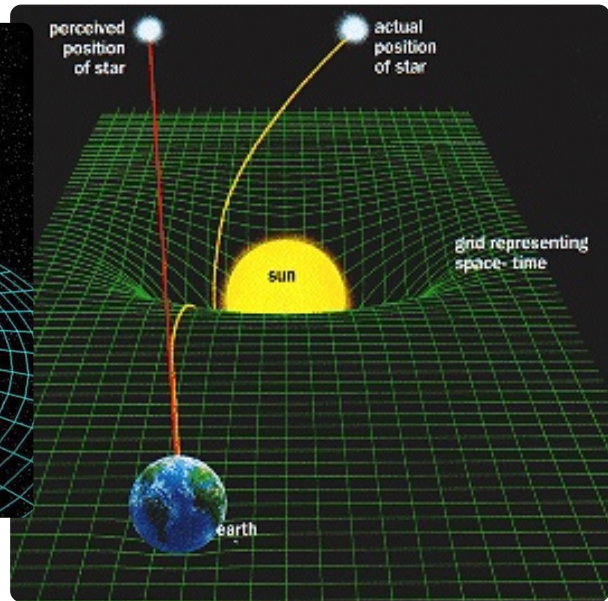
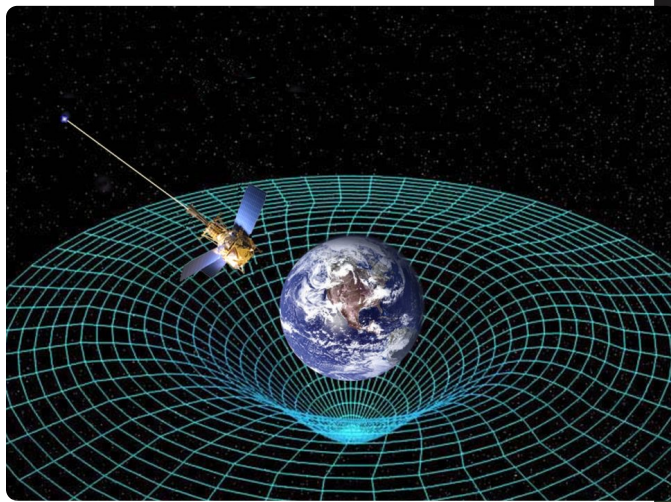


← beam of light being pulled by gravity.

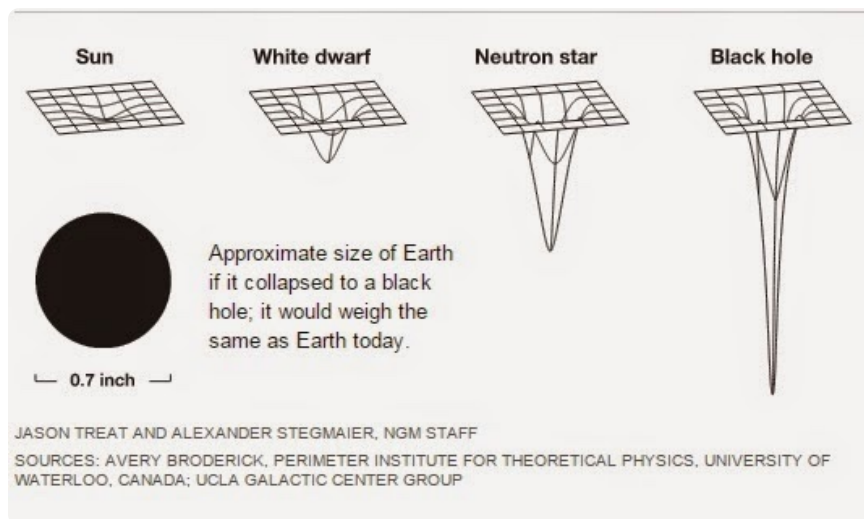
Einstein concluded that what we gravity is actually 4D spacetime that's curved. (13)

Mass causes spacetime to bend.

The more heavy an object the more it bends space



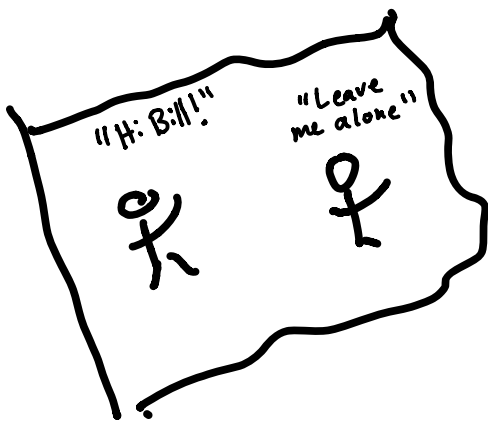
The smaller the space with the same mass curves spacetime more.



We don't see space curved. ⁽¹⁴⁾

This is because we only see in 3D

An Example: Say we live in the 2D paper world.

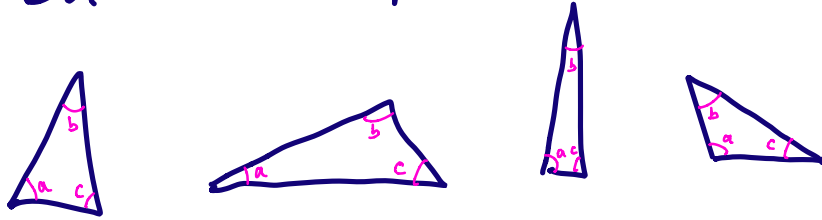


If we bend/curve the paper in the 3rd dimension the paper people wouldn't notice.

However we can use geometry to show our space is intrinsically curved.

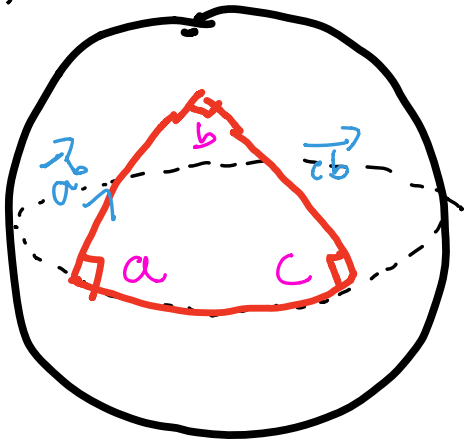
One property of curved surfaces is (15)
that the sum of angles of a triangle
 $\neq 180^\circ$

In flat space



For all of these $a + b + c = 180^\circ$

Now take a ball: I can draw 3
 90° angles and
get a triangle



$$a + b + c = 270!$$

Try it yourself!

This also means two parallel lines
will intersect.

We can apply these properties to
objects falling in gravity to calculate the
curvature of space.

5.) We saw how $\text{sum of angles} \neq 180^\circ$ in curved space time. (16)

How does Curve space change other math?

Let's start by looking at how we describe the expansion of our universe

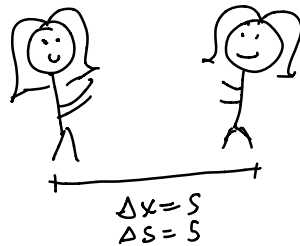
recall: $(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c\Delta t)^2$

We can add an "expansion" $a(t) = t$

$$(\Delta S)^2 = a(t) \left((\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 \right) - (c\Delta t)^2$$

at $t=1$ we get back our flat geometry

let two friends be separated by $\Delta x = 5$ at $t=1$, $a(t)=1$



We find their distance is $(\Delta S)^2 = 1 \cdot (\Delta x)^2 = 25$

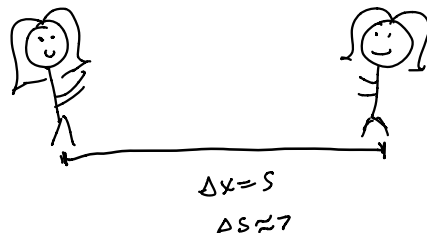
at $t=2$ $a(t)=2$ and $(\Delta S)^2 = 2(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c\Delta t)^2$

Without moving the space between the twins grew.

$$(\Delta S)^2 = 2(\Delta x)^2$$

$$= 2(25) = 50$$

$$\sqrt{(\Delta S)^2} = \sqrt{50} \Rightarrow \Delta S = 7.07$$



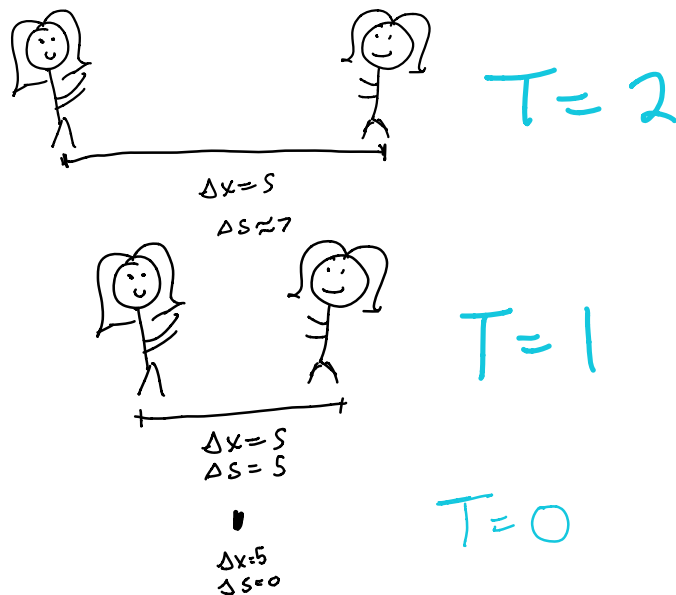
Big Bang.

⑦

Take $a(t) = t$ at $t=0$

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c\Delta t)^2 = -(c\Delta t)^2$$

This model suggests the distance between the friends at $t=0$ is $\Delta S=0$



Thus, we found a geometry that includes the big bang. We considered $a(t) = t$. Our universe is not just expanding it is accelerating. If you have taken highschool physics than you'll know that $a(t) \sim t^2$ gives acceleration.

Now that we seen a possible correction (18)
 Our "distance equation" (aka Metric):

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c \Delta t)^2$$

Lets see how we can change it to
 include the effects of gravity:

Roughly this looks like.

$$(\Delta S)^2 = \frac{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}{\left(1 - \frac{GM}{c^2 r}\right)} - \left(1 - \frac{GM}{c^2 r}\right) (c \Delta t)^2$$

This is a curved metric. Using this
 to calculate your motion falling to earth
 produces the familiar Newton gravitational potential:

$$V = -\frac{GM}{r}$$

Which is a factor found in
 our new equation

We can check as we move far away from
 earth: $V = -\frac{GM}{\infty} = 0$ The potential vanishes.

$$(\Delta S)^2 = \frac{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}{\left(1 - \frac{GM}{c^2 \infty}\right)} - \left(1 - \frac{GM}{c^2 \infty}\right) (c \Delta t)^2$$

$$= (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c \Delta t)^2$$

When we move very far away we return to flat space

Thus the further we move away from earth, the more flat our geometry becomes and the less we feel the pull of gravity. (19)

If you have taken pre-calc then expressing this in terms of r, θ, ϕ is proper since the earth is a sphere.



This is called Schwarzschild Metric:

$$(\Delta S)^2 = -\left(1 - \frac{GM}{c^2 r}\right) (c \Delta t)^2 + \frac{(\Delta r)^2}{\left(1 - \frac{GM}{c^2 r}\right)} + r^2 (\Delta \theta)^2 + \sin^2 \theta (\Delta \phi)^2$$

Aside

Where on a sphere: $x = r \sin \theta \cos \phi$ $y = r \sin \theta \sin \phi$ $z = r \cos \theta$

What happens when all the mass shrinks down below $r = \frac{GM}{c^2}$?

The earth is much larger than this area but if we were at $r = \frac{GM}{c^2}$ our equation says

$$(\Delta S)^2 = 0 (c \Delta t)^2 + \infty (\Delta r)^2$$

This means we are at an event horizon of a black hole where time appears to freeze.

Black Holes

(20)

Before we talk about what happens at the event horizon, let's see their

Size:

Earth:



Mass = $5.972 \times 10^{24} \text{ kg}$
radius to earth's center = 3,458 miles

The radius the earth must shrink to to become a black hole

$$r_{\text{event horizon}} = \frac{G (5.972 \times 10^{24} \text{ kg})}{c^2}$$

$$G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

$$c = 299\,792\,458 \text{ m/s}$$

$$r_{\text{event horizon}} = .4 \text{ cm}!$$

The Earth must shrink $\frac{1}{4}$ the size of a dime to be a black hole!

Kanye!



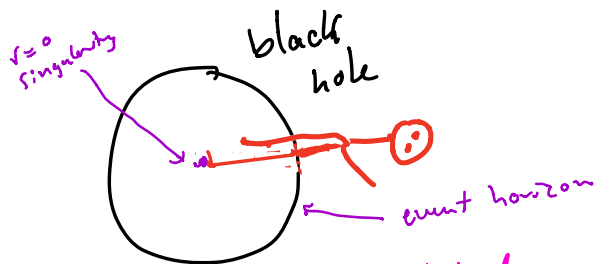
Mass = 75 kg

(21)

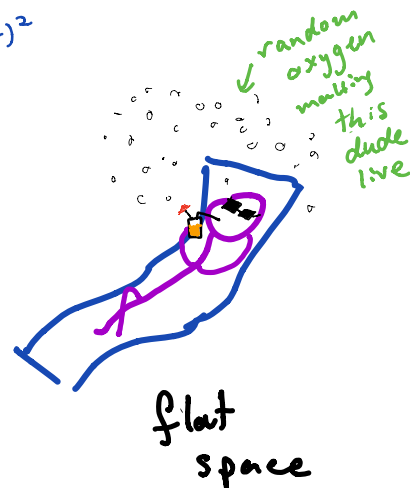
$r_{\text{event horizon}} = 5.5 \times 10^{-26} \text{ cm}$

As stated, someone looking at the black hole from the outside will see time "stops" at the event horizon and will never see the person cross.

$$\Delta s^2 = 0(\Delta t)^2 + \infty(\Delta r)^2$$



Poor person being stretched entering a black hole since gravity is stronger at the feet than the head.



The person falling into a black hole will do so in a short amount of time

~ 6 microseconds to go from horizon to singularity.

In the fraction of a second it takes
for the person to fall into the black hole,

(22)

INFINITE amount of time passes for the
person chillin outside.

This is one of the reasons why **Nothing**
Can escape a black hole!

In order to escape a black hole you
must go faster than light. Which
requires an infinite amount of energy and
is not possible.

Light cannot escape a black hole

We can see this with a diagram.
In flat space light moves at 45° .



The light rays going out are fighting curved space and take longer time to reach a person in flat space.

(23)

At the event horizon outgoing light rays are trapped! They don't leave the event horizon. (Their fighting exactly cancels the pull of gravity or curvature of space)

All Light rays inside the event horizon, will always go to the singularity $r=0$.

Every time mass falls into the event horizon the Area grows.

As we saw $r_{\text{Event horizon}} = \frac{2GM}{c^2}$

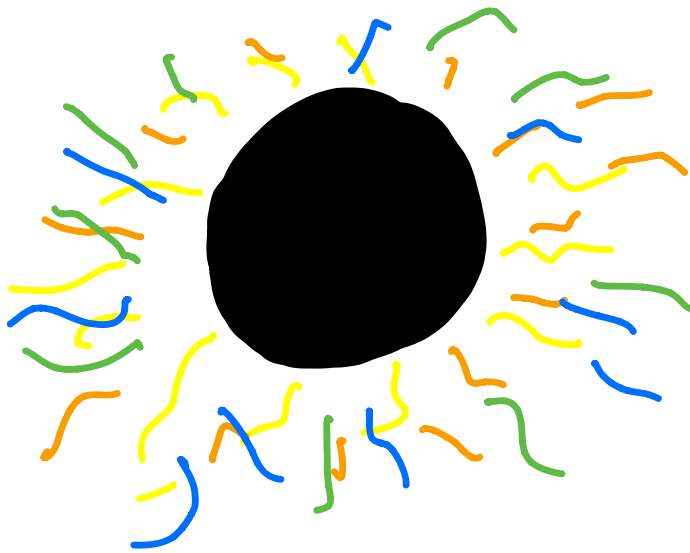
But can it shrink?

The answer is yes!

Hawking Radiation

(24)

Black holes, just like everything else, radiate.



It can radiate all different things

EX:

- Light
- Electrons
- parts of protons (quarks)

This is bizarre considering a black hole is just space itself.

An explanation involves quantum mechanics and deals with the fact particles pop in and out of existence.

You can type "Hawking Radiation" into Youtube to find out more.

(25)

aside

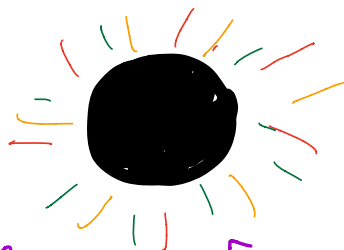
The hotter things are the more they radiate.

The temperature of a black hole is.

temp $\rightarrow T = \frac{1}{8\pi M} \leftarrow \text{mass}$

The time it takes: $t = M^3 \frac{5120\pi G^2}{h c^4}$
 $C = 6.67 \times 10^{-11}$
 $C = 3 \times 10^8$
 $h = 6.6 \times 10^{-34}$

Therefore big massive blackholes are colder and radiate less than lighter smaller black holes.



↑ Smaller blackhole loses mass faster than a bigger more stable one.

The time it will take for a black hole the mass of the earth to evaporate is 10^{67} years.
 One the size of kanye is 2.2×10^{-40} seconds
 Big Difference!

Can a black hole radiate away to nothing?

Quantum Gravity / String Theory (26)

The issue with answering this question is we don't know what space means at distances less than $3 \times 10^{-35} \text{ m}$.

Using Einstein's theory there is no way to talk about a black hole smaller than $3 \times 10^{-35} \text{ m}$,

This is called the plank length.

Some theories suggest there is no space at all.

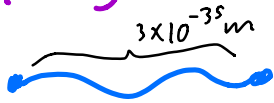
String theory provides an interesting answer to what happens when a black hole radiates to be smaller than the plank length.

String Theory says all the particles we think are points are actually small strings the size of plank length.

(27)

●
↑
electron.

They can be open or closed.



The more they vibrate the heavier they are.

As a black hole shrinks radiating particles (**strings**), its event horizon reaches the size of about the plank length and it just turns to a highly

Vibrating

String :



(or in your concept
a very heavy particle)

Aside:

String Theory also suggests the universe is a hologram. Just like how the event horizon is a 2D sphere covering a 3D black hole. Our 3D space might be governed by a 2D surface surrounding our universe

Check out "The holographic principle" on youtube or online